

# Disproof of TLMC Conjecture 00000001211: P-Positions of the Young Poset Game Are Not the Perfect Squares

## Conjecture and verdict

**Conjecture 1** (TLMC 00000001211, verbatim). *Definition: A poset game (delete an element and everything above it). Conjecture: The P-positions of the poset game on Young’s lattice have an explicit characterization via perfect square partitions (the Young poset game).*

**Verdict: disproved.** The phrase “poset game on Young’s lattice” admits two natural readings; under *both* the perfect-square characterization fails, and every number in this note is certified twice: by a Lean development that uses *no axioms at all* (no sorry, no `Classical.choice`, no `Quot.sound`, not even `propext`; see `lean4/Check.lean`) and by an independent Python enumeration over all 252 Young diagrams fitting in the  $5 \times 5$  box (`reproduce.py`).

## Setup

A *Young diagram* is a partition  $\lambda = (\lambda_1 \geq \lambda_2 \geq \dots \geq 0)$ , drawn as rows of cells, row  $i$  (1-indexed, from the top) having  $\lambda_i$  cells. A move *chomps* a cell  $(i, j)$ : the cell together with every cell  $(r, c)$  with  $r \geq i, c \geq j$  is removed. Under normal play, a position is a *P-position* if the player to move loses. A *perfect-square partition* is  $(k, k, \dots, k)$  with  $k$  rows of length  $k$  (a  $k \times k$  square).

**Reading 2** (Literal poset game). Every cell of the current diagram may be chomped; the player who cannot move (facing  $\emptyset$ ) loses.

Since the corner cell  $(1, 1)$  belongs to every nonempty diagram and its up-set is the whole diagram, chomping  $(1, 1)$  clears the board in one move.

**Theorem 3** (Reading 1). *The P-positions of the literal Young poset game are exactly  $\{\emptyset\}$ . In particular the perfect square  $(2, 2)$  is an N-position.*

*Lean certificate.* `isPLit_nil : isPLit [] = true` and `isPLit_cons : 1 <= n -> isPLit (n :: rest) = false`, i.e. no nonempty position is a P-position; combined they pin  $P = \{\emptyset\}$ . The key lemma `chompAt_11` states that chomping  $(1, 1)$  always yields the empty diagram; the witness move then makes every nonempty position an N-position.  $\square$

**Counterexample 4** (Reading 1).  $(2, 2)$  is a perfect-square partition but an N-position (indeed so is every nonempty diagram). If one insists on regarding  $\emptyset$  as the  $0 \times 0$  square, the inclusion “ $P \subseteq$  squares” holds trivially while “squares  $\subseteq P$ ” fails; either way the explicit characterization is false.

**Reading 5** (Chomp with a poison corner). The corner  $(1, 1)$  is poison: taking it loses. Equivalently—since a rational player never takes poison voluntarily—forbid that move; a player facing the lone poison cell  $(1)$  has no winning move and must take the poison, so  $(1)$  is a P-position, exactly as in Chomp.

**Theorem 6** (Reading 2). *The characterization fails in both directions:  $(2, 1)$  is a  $P$ -position that is not a square, and the perfect squares  $(2, 2)$  and  $(3, 3, 3)$  are  $N$ -positions.*

*Lean certificate.* `chomp_counterexample` bundles the `decide`-verified evaluations `isPChomp [2,1] = true, isSquare [2,1] = false, isSquare [2,2] = true, isPChomp [2,2] = false, isSquare [3,3,3] = true, isNChomp [3,3,3] = true`.  $\square$

## The $5 \times 5$ enumeration

There are  $\binom{10}{5} = 252$  partitions fitting in the  $5 \times 5$  box. The literal reading yields the single  $P$ -position  $\emptyset$ . The Chomp reading yields exactly 24  $P$ -positions:

$(\emptyset), (1), (2, 1), (2, 2, 1), (3, 1, 1), (3, 2), (2, 2, 2, 1), (4, 1, 1, 1), (4, 3), (3, 3, 1, 1), (4, 2, 2),$   
 $(2, 2, 2, 2, 1), (4, 2, 1, 1, 1), (5, 1, 1, 1, 1), (5, 2, 1, 1), (5, 4), (3, 3, 2, 1, 1), (5, 3, 2), (3, 3, 3, 2, 2),$   
 $(4, 4, 3, 1, 1), (5, 3, 3, 2), (5, 5, 3), (4, 4, 2, 2, 2), (5, 5, 2, 2).$

The only square among them is the degenerate  $(1)$ ; the nondegenerate squares  $(2, 2), (3, 3, 3), (4, 4, 4, 4), (5, 5, 5, 5, 5)$  are all  $N$ -positions (as is the  $2 \times 3$  rectangle  $(3, 3)$ ).

The  $3 \times 3$  corner of the table, under both readings:

| position    | Reading 1 (literal) | Reading 2 (Chomp)       |
|-------------|---------------------|-------------------------|
| $\emptyset$ | P                   | — (not reached)         |
| $(1)$       | N                   | P (poison)              |
| $(2)$       | N                   | N                       |
| $(1, 1)$    | N                   | N                       |
| $(2, 1)$    | N                   | <b>P</b> (not a square) |
| $(2, 2)$    | N                   | <b>N</b> (a square)     |

For instance, under Reading 2 the moves of  $(2, 1)$  chomping cells  $\neq (1, 1)$  are  $(1, 1)$  and  $(2)$ , both  $N$  (each has the move to  $(1)$ ), so  $(2, 1)$  is  $P$ ; then  $(2, 2)$  has the move  $(2, 2) \rightarrow (2, 1)$  to a  $P$ -position and is  $N$ .

## Machine-checked certificate

`lean4/Main.lean` encodes diagrams as `List Nat` (non-increasing row lengths) and defines the chomp move `chompAt`, the cell enumeration `anyCell/cols`, and two fuel-based win oracles `winLit/winChomp` with the  $N/P$  predicates `isNLit/isPLit/isNChomp/isPChomp` at fuel `total( $\lambda$ ) + 1`. The fuel is sufficient: every move enumerated is a genuine cell (column  $j$  of a row of length  $n$  has  $1 \leq j \leq n$ ), hence row  $i$  shrinks from  $len$  to  $\min(len, j - 1) \leq len - 1$  and every move removes at least one cell, so the game depth from a diagram with  $t$  cells is at most  $t$ . Chomped rows that become empty end the construction immediately (for genuine cells of canonical partitions this can only happen when  $j = 1$ , which empties all lower rows too), so results stay canonical partitions; this also keeps every definition free of `propext`.

`lean4/Check.lean` prints `#print axioms` for all 18 theorems; every line reads “*does not depend on any axioms*”. `reproduce.py` re-derives the enumeration independently and asserts every figure quoted above.

## Conclusion

Conjecture 00000001211 is false under both readings of the game: the P-positions of the Young poset game are  $\{\emptyset\}$  (literal reading) or a 24-element set inside the  $5 \times 5$  box that neither contains nor is contained in the set of perfect-square partitions (Chomp reading). No perfect-square characterization of the conjectured form exists.